

# Homework 3 – Mesh Generation for CFD Analysis

## Submission

| Page  | Description                                                      |
|-------|------------------------------------------------------------------|
| 1     | Manifold - A snapshot of the overall mesh                        |
| 2     | Manifold - A snapshot of the mesh interior using a section plane |
| 3     | Manifold - An example mesh which is much finer                   |
| 4,5,6 | Tasks from the tutorials for T-Junction and heat mixer           |

## Manifold Mesh Generation

The default meshing method was used with capture curvature enabled and a global element size of 1.5 mm. Inflation layers were introduced near the walls to capture boundary layer effects. The mesh consists mainly of triangular elements distributed across the computational domain. Inflation settings:

- Number of layers: 10
- Growth rate: 1.2
- Maximum thickness: 1.5 mm

## Mesh Quality Evaluation

Mesh quality was evaluated using the orthogonal quality metric in ANSYS Meshing. The minimum orthogonal quality of 0.1807 falls in the acceptable range, indicating that the lowest-quality cells are still usable for CFD simulations. The average value of 0.7018 lies in the very good range, showing that most mesh elements have good geometric quality. The maximum value of 0.9879 indicates that some elements are close to the ideal cell shape.

Overall, the mesh quality is good and suitable for CFD analysis. The addition of inflation layers near the walls improves the resolution of the boundary layer and supports accurate flow predictions.

### Orthogonal Quality Metrics

| Metric             | Value  |
|--------------------|--------|
| Minimum            | 0.1807 |
| Maximum            | 0.9879 |
| Average            | 0.7018 |
| Standard deviation | 0.1458 |





Context A : Fluid Flow (Fluent) - Meshing [CFD PrepPost, CFD PrepPost Pro, CFD Base]

File Home Mesh Display Selection Automation Learning and Support

Quick Launch

Tools

Outline

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Project\*

Model (A3)

- Geometry Imports
- Geometry
- Materials
- Coordinate Systems

Details of "Mesh"

|                             |                            |
|-----------------------------|----------------------------|
| Physics Preference          | CFD                        |
| Solver Preference           | Fluent                     |
| Element Size                | 1.5e-003 m                 |
| Export Format               | Standard                   |
| Export Preview Surface Mesh | No                         |
| Sizing                      |                            |
| Use Adaptive Sizing         | No                         |
| Growth Rate                 | Default (1.2)              |
| Max Size                    | 1.5e-003 m                 |
| Mesh Defeaturing            | Yes                        |
| Defeature Size              | Default (7.5e-006 m)       |
| Capture Curvature           | Yes                        |
| Curvature Min Size          | 1.5e-004 m                 |
| Curvature Normal Angle      | 18.0°                      |
| Capture Proximity           | No                         |
| Bounding Box Diagonal       | 8.6079e-002 m              |
| Average Surface Area        | 3.8108e-004 m <sup>2</sup> |
| Minimum Edge Length         | 8.255e-003 m               |
| Quality                     |                            |
| Check Mesh Quality          | Yes, Errors                |
| Target Skewness             | Default (0.9)              |
| Smoothing                   | High                       |
| Mesh Metric                 | Orthogonal Quality         |
| Min                         | 0.18068                    |
| Max                         | 0.98788                    |
| Average                     | 0.70182                    |
| Standard Deviation          | 0.14579                    |
| Inflation                   |                            |
| Use Automatic Inflation     | Program Controlled         |
| Inflation Option            | Total Thickness            |
| Number of Layers            | 10                         |
| Growth Rate                 | 1.2                        |
| Maximum Thickness           | 1.5e-003 m                 |

Ready

Ansys 2025 R2 STUDENT

0 0.01 0.02 (m)

0.005 0.015

Mesh Metrics

Controls

Number of Elements

Element Metrics

Tet4 Wed6

11457.00

0.18 0.25 0.38 0.50 0.62 0.75 0.88 0.99

No Messages No Selection Metric (m, kg, N, s, V, A) Degrees rad/s Celsius

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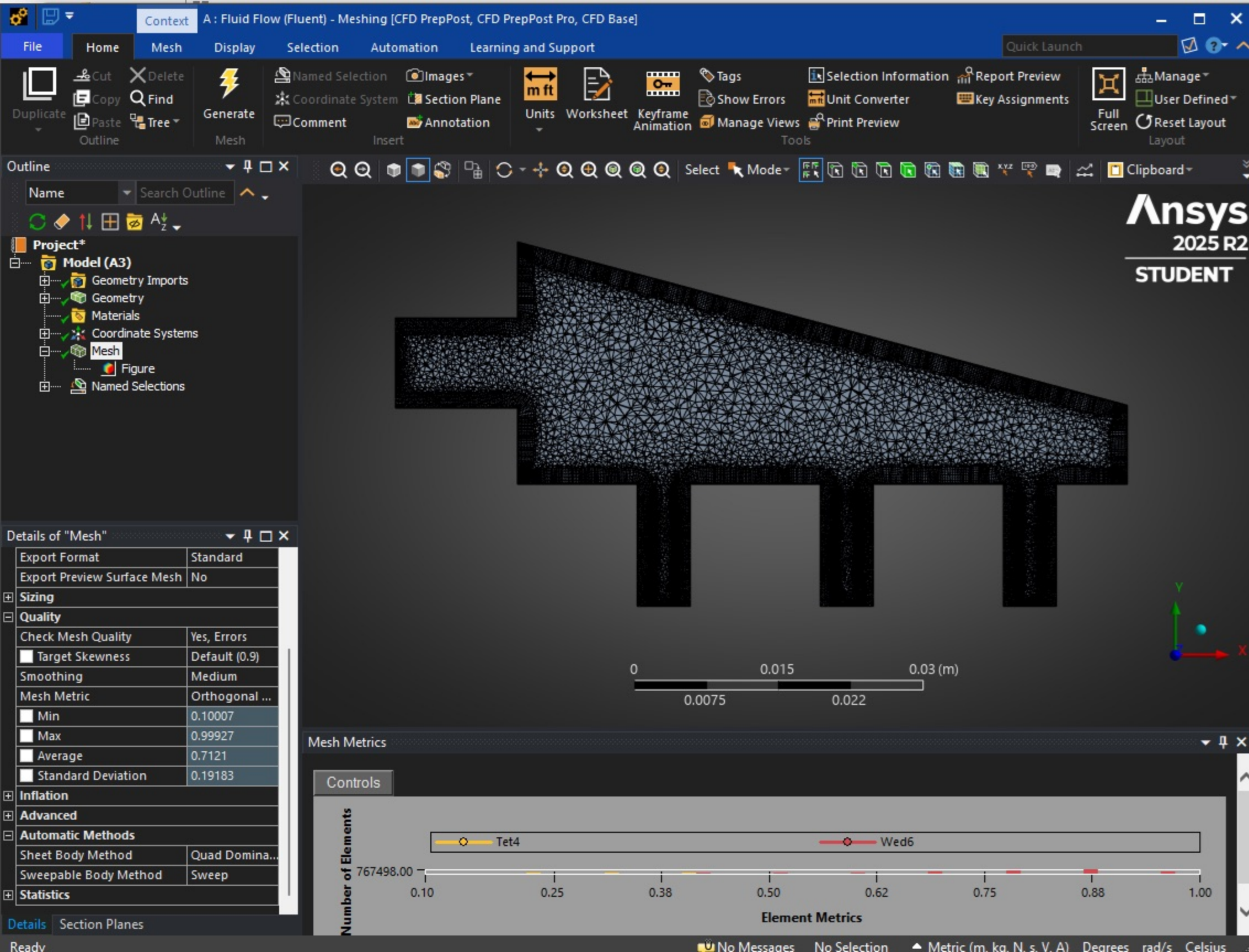
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